CMSC 27100-3 Due: Sunday, October 18, 2020

Problem Set

Instructor: Robert Rand

1 Set Theory

Problem 1: Solve the following equations:

1. $\{2,4,6,8\} \cup \{1,2,3\}$
2. $\{2,4,6,8\} \cap \{1,2,3\}$
3. $\{2,4,6,8\} \setminus \{1,2,3\}$
4. $\{x \in \mathbb{N} \mid \exists y, x = 2y\} \cup \{x \in \mathbb{N} \mid \exists y, x = 2y+1\}$
5. $\{x \in \mathbb{N} \mid \exists y, x = 2y\} \cap \{x \in \mathbb{N} \mid \exists y, x = 2y+1\}$

Problem 2: True or False?

1. $2 \in \{2,4,6,8\}$
2. $4 \subset \{2,4,6,8\}$
3. $\{6\} \subset \{2,4,6,8\}$
4. $8 \in \{x \in \mathbb{N} \mid \exists y, x = 2y\}$
5. $\{2,9\} \subseteq \{x \in \mathbb{N} \mid \exists y, x = 2y\}$

Problem 3: Write out the set of prime numbers using set-builder notation (aka set comprehension). You may use standard mathematical symbols, but may not use the predicate Is_Prime (or equivalent).